Class 3

How do water flows interact with a bed made of sand?

What a strange, strange boy
He sees the cars as sets of waves
Sequences of mass and space

Joni Mitchell,
'A strange Boy'
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Background to the practical

So far in the course we have looked at the key physical properties of sediment and fluids, and thought about how these interact to affect the movement of sediment grains. In this practical class we will be using a flume (an open-topped, man-made channel) to observe how water flows interact with a bed made of sand and produces some of the bedforms and sedimentary structures we will examine next week.

The aim of this experiment is to explain what happens to a sand bed as a unidirectional water flow above it is increased in its capacity. This week we will focus on making qualitative and quantitative observations.

We will conduct one experiment. In our experiment we will be running an unidirectional (one-way) water flow over a sand bed. The grain size of the sediment will be held constant (mean=255 μm; range 300-150 μm), as will the grain density (quartz), fluid density and viscosity (water @20°C).

Intended learning outcomes

At the end of this practical class you should be able to:

- use laboratory equipment to measure and record key variables during the experiment including flow depth, velocity, structure, bed slope, bedform migration rate
- illustrate your observations using interpreted, annotated diagrams which show both the physical observations (e.g. bedform and sedimentary structure) and processes of erosion, transport and deposition
- relate changes in the way in which particles move, the sediment transport rate and bedform created to changes in the flow strength
- work co-operatively in a research team to follow a experimental procedure to successfully complete the experiment in the time available. This will require you to make evaluate your observations during the experiment and make adjustments to the instructions as required.

SAFETY NOTE: WATER AND ELECTRICITY ARE A LETHAL COMBINATION!!!!

Experiment 1: Bedforms in fine to medium grain sand

Objective

The objective of this experiment is to demonstrate the sequence bedforms which appear under an unidirectional open channel flow, with a mobile sand-bed, as the intensity of sediment transport increases.

Equipment Required

Flume (demonstration channel)
Stop watch
Digital spirit level
Ruler
Neutrally buoyant green balls
Velocity meter
Chinagraph pencil
Tape measure
Red food colouring and pipette

Equipment Set Up

Sand of a uniform size between 0.1 and 0.3mm is placed in the flume to a uniform depth

The sand surface is tamped down so that it is uniformly packed, lightly compacted and as flat as possible. The flow valve is fully closed. The flume slope is set to zero.

You will work in research teams of 8 members. You will need to divide the following control and measurement tasks among your team members before you begin:

Job 1: the fat controller (in charge of the flume controls) and flood watch
Job 2: data recorder
Job 3: measure bed slope (using digital spirit level)
Job 4: flow visualiser (using red dye and pipette)
Job 5: bedform migration rate (using chinagraph pencil, stop watch and ruler)
Job 6: flow velocity Method 1 (using velocity meter)
Job 7: flow velocity Method 2 (using a stop watch and neutrally buoyant green ball)
Job 8: flow depth (ruler)

Everyone should make their own observations and calculations

Preparation

Mark the central 50cm of the flume on the glass using the chinagraph pencil.

All observations and measurements should be made in within this working reach of the flume (the central 50 cm) and well away from any entrance or exit effects.

Mean depth should be based on at least 5 point measurements of water depth,10cm apart within the working reach. Note that the water depth is measured from the free surface to the top of the sand bed, not to the solid flume bed.

Make sure you know how to use the piece of equipment you are responsible for.

Procedure

At each stage of the experiment be patient and allow the flow and bed to adjust to the new settings (i.e. reach some sort of equilibrium). At a minimum, allow the new bed state to be established through the middle working section of the flume.

Ensure that the bed slope is set to a zero slope and open the flow control valve two turns. Turn on the pump, making sure that the pump is set to its lowest setting I, and allow the water to enter the flume. Allow the tank to setting a couple of minutes.

Under these conditions the bed should be immobile. Wait until steady flow along the length of the flume is achieved, and check that there is no bed motion.

To be successful, it is important to observe the condition of the bed closely and carefully during the rest of the experiment. **Increase the slope a fraction to a non-zero value**, but not enough to exceed the threshold of bed material motion. To find this point you will need to decide how many grains need to be moving for the threshold of motion to be exceeded. Before the threshold is exceeded there is *(almost)* zero sediment transport and the bed is immobile. Under these conditions there are no bedforms and the configuration of the bed is that of a plane bed with no motion.

A1. Basic description of the flow and bed just prior to the initiation of sediment transport

Is the flow structure laminar or turbulent?
What two methods could you use to discover this?

- Visualise the flow structure by dropping some red dye into the flow- does it trace out a line or is it mixed?

  Answer:
- Measure the velocity and water depth and calculate the flow Re

  Answer:

The results should demonstrate that all flows in the flume, and also in natural rivers, are fully turbulent even when the surface of the water is smooth and glassy.

A2. Conditions at the initiation of sediment transport: Forces exerted on the bed

What was the critical shear stress (τ_{cr}) needed to get the grains to move?

Remember for grains to move τ_0 (the applied force on the bed by the flow) = τ_{cr}

Using the flume take the bed to just after the threshold of motion.

Increase the slope incrementally until there is a small but finite sediment load. Well done- you have initiated bedload transport. Almost immediately small, triangular bedforms will appear. These are **ripples.** They are a few mms high and a cm or two long and are asymmetrical, with a gentle upstream slope and a steep downstream face. Observe the motion of grains making up the ripples. Observe the geometry of the water surface.

- Record the mean flow depth, slope and velocity.

You can calculate the shear stress at the initiation of transport (when ripples started to form) using Du Buoys equation:

$$\tau_0 = \rho g h S$$

where:
τ_0 = shear stress acting on the bed,
ρ = density of water
S = slope (m/m or $\sin\alpha$), and
g = acceleration due to gravity

Answer: τ_0 =

A3. Transition to dunes

Increase the pump discharge to setting III. The flow depth and rate of sediment movement will both be seen to increase immediately. **If necessary, increase the slope incrementally** until grains are seen to hop and bounce periodically, even though the predominant mode of motion is still by rolling. This motion by hopping or bouncing is referred to as **saltation.**

Under this flow intensity, larger triangular bedforms called **dunes** become dominant. The size of dunes produced in a small flume is not much greater than that of ripples in nature. However, they are also longer and flatter topped, and disturb the free surface of the flow.
Once the dune bed has been established throughout the working section of the flume, observe that the water surface is drawn down slightly over a dune crest, but is humped over the trough between dunes. The waves in the water surface are thus out of phase with the waves in the bed. Note the presence of any **boils** associated with turbulence in the lee of the dune. Observe the process of sediment movement downstream, with **rolling and saltation** of grains up the stoss slope. Observe grains **avalanching** down the lee slope, and watch for the presence of **separation bubbles** on the lee slopes, causing upstream flow which sweeps grains back up the face. Observe the orientation of the dunes in the dune bed, the pattern of dune crests, and note the presence and position of any **scour holes.**

- In the space below, draw a careful sketch of one dune in cross-section, from the trough of one dune to the trough of the next.
    - Include a scale: dune height and wavelength
    - Annotate the dune profile with the names of the different parts of the dune geometry
    - Use arrows to show the local flow directions, and label the points of flow separation flow reattachment, and separation vortex
    - Show on the dune profile where particles are eroded, changes to how grains are transported, and where deposition is occurring
    - Interpret from your observations changes in flow velocity and bed shear stress over the dune profile and annotate the diagram with these

- Record the migration rate of the dunes down flow using the chinagraph pencil, a stopwatch and ruler.
  Answer:

- Measure a mean flow velocity using Method 1
  Answer:

- Measure the surface velocity using Method 2
  Answer:

- Compare your results- why are they so different?

A4. Dune planform and flow strength

Open the flow valve fully so that you are producing maximum discharge. The flow depth again increases, as does the intensity of sediment motion. Saltation will become more pronounced and the movement of some grains as **suspended load** maybe observed, as grains leap from the crest of one dune directly onto the stoss slope of the next without coming to rest in the trough.

- Observe changes in the amount of sediment transported and the way particles are moving. How does this relate to changes in the **dune** height and wavelength you can see?

A5. Transition from dunes to upper plane bed

Now **make a substantial increase in slope**, and observe that the dunes lengthen and reduce in amplitude. The resulting long, low bedforms are called **washed-out dunes.** Note their shape and the presence (or lack of) troughs and scour holes. Note the bedload motion occurring, and the presence or absence **of boils and drawdown areas** in the water surface.

When the wash-out dunes occupy the whole working length, record the flow velocity, mean water depth and flume slope.

Time is of the essence in the remaining parts of the experiment because the high rates of bed material transport associated with the next bedforms will over whelm the capacity of the flume to recirculate sediment, resulting in rapid scour down to the solid flume bed. Measurements must be made as quickly as possible once each bedform regime has become established.

Make a further substantial increase in flume slope. Observe the disappearance of the washed-out dunes and the establishment of a **plane bed with motion.**

- Observe changes in the water depth and the appearance of the water surface, and the presence or absence of boils and surface turbulence.
- Observe the appearance of the moving bedload and from above note the presence of ‘streaks’ along the flume bed parallel to the flow direction.

A6. Conditions during the formation of standing waves and antidunes

Increase the flume slope by a small amount.

- Observe that the flat, glassy smooth water surface begins to deform into almost stationary undulations called **standing waves.**
- Observe the appearance of the flume bed, and note any relationship between the bed features and the flow surface.

Make a fairly substantial increase in flume slope to move the flow well into the upper regime.

- Observe the amplitude of the surface waves and the bedforms below them, and note the symmetry and regularity of the bedforms, which are known as **antidunes.**
- Observe the motion of grains as they are eroded from the downstream face of one antidune and deposited on the upstream face of the next antidune downstream.

Motion of the antidune moves in the upstream direction, against the movement of the water flow and the grain motion. This could be verified by tracing the outline of an antidune on the flume wall using a chinagraph pencil and observing its movement through time. It must be recognised that the water and the grains move downstream at all times, even though the bedform moves upstream.

Increase the flume slope a moderate amount. This produces a corresponding increase in the intensity of sediment motion. The rate of antidune movement increases. An important new development is that as the amplitude of the antidunes increase, the energy loss involved in flow up the slope to the crest becomes progressively greater. There comes a point when the flow at the crest appears to stall and intense turbulence occurs. This destroys the bedform, which breaks and dissipates in the upstream direction. These bedforms are called **breaking-antidunes.** Usually, the breaking is not random because breaking of a downstream antidune triggers breaking of the next wave upstream, and so on along the length of the flume, in nature this leads to pulsing flow as trains of antidunes form and break, although such forms are relatively rare due to the high and sustained transport of bed sediment involved.